

C# Coding Style Cheat Sheet

File Organization

1. **Using** statements (System first, **then** alphabetical)
2. **Namespace** declaration
3. **Class/Interface** declaration
 - ├ Constants
 - ├ **Static** fields
 - ├ Instance fields
 - ├ Constructors
 - ├ Properties
 - ├ **Public** methods
 - ├ **Private** methods
 - └ Nested types

Naming Conventions

Element	Convention	Example
Classes	PascalCase	CustomerOrder
Interfaces	IPascalCase	IRepository
Methods	PascalCase	CalculateTotal()
Properties	PascalCase	FirstName
Local variables	camelCase	orderCount
Parameters	camelCase	customerId
Private fields	_camelCase	_connectionString
Constants	UPPER_CASE	MAX_RETRY_COUNT
Enums	PascalCase	OrderStatus.Pending

Indentation Rules

- **Use spaces, not tabs** (4 spaces per level)
- **Braces on new lines** (Allman style)
- **One statement per line**
- **Blank line between logical sections**

```
// ✓ Correct
public void ProcessOrder(Order order)
{
    if (order == null)
    {
        throw new ArgumentNullException(nameof(order));
    }

    ValidateOrder(order);
    SaveOrder(order);
}

// x Wrong
public void ProcessOrder(Order order) {
    if (order == null) { throw new ArgumentNullException(nameof(order)); }
    ValidateOrder(order); SaveOrder(order);
}
```

Statement Guidelines

If Statements

```
// Always use braces, even for single lines
if (condition)
{
    DoSomething();
}

// Avoid nested ternary
// x Bad: result = a ? b ? c : d : e;
// ✓ Good: Use if-else for complex logic
```

Switch Statements

```
switch (status)
{
    case Status.Active:
        ProcessActive();
        break;
    case Status.Pending:
        ProcessPending();
        break;
    default:
        throw new InvalidOperationException();
}
```

LINQ

```
// Multi-line for readability
var results = customers
    .Where(c => c.IsActive)
    .OrderBy(c => c.Name)
    .Select(c => new { c.Id, c.Name })
    .ToList();
```

White Space Rules

Context	Rule
After keywords	if (, for (, while (
Around operators	x = y + z
After commas	Method(a, b, c)
No space before (in methods	DoSomething()
Blank line between methods	Always
No trailing whitespace	Never

Quick Checks Before Commit

- ☐ File organization follows standard order
- ☐ All names follow conventions
- ☐ Consistent indentation (4 spaces)
- ☐ Braces on new lines
- ☐ No trailing whitespace
- ☐ Blank lines between logical sections
- ☐ No commented-out code
- ☐ No magic numbers (use constants)

The Golden Rule

“Code is read 10x more than it’s written. Write for the reader.”

Generated from Ask Marilyn About Software Testing - C# Coding Style Guide Course